Data Science with Python

A six-week hands-on workshop

From Basics to Machine Learning

Shadab Tabatabaeian

Postdoctoral Researcher

Senior Data Scientist

Georgetown University

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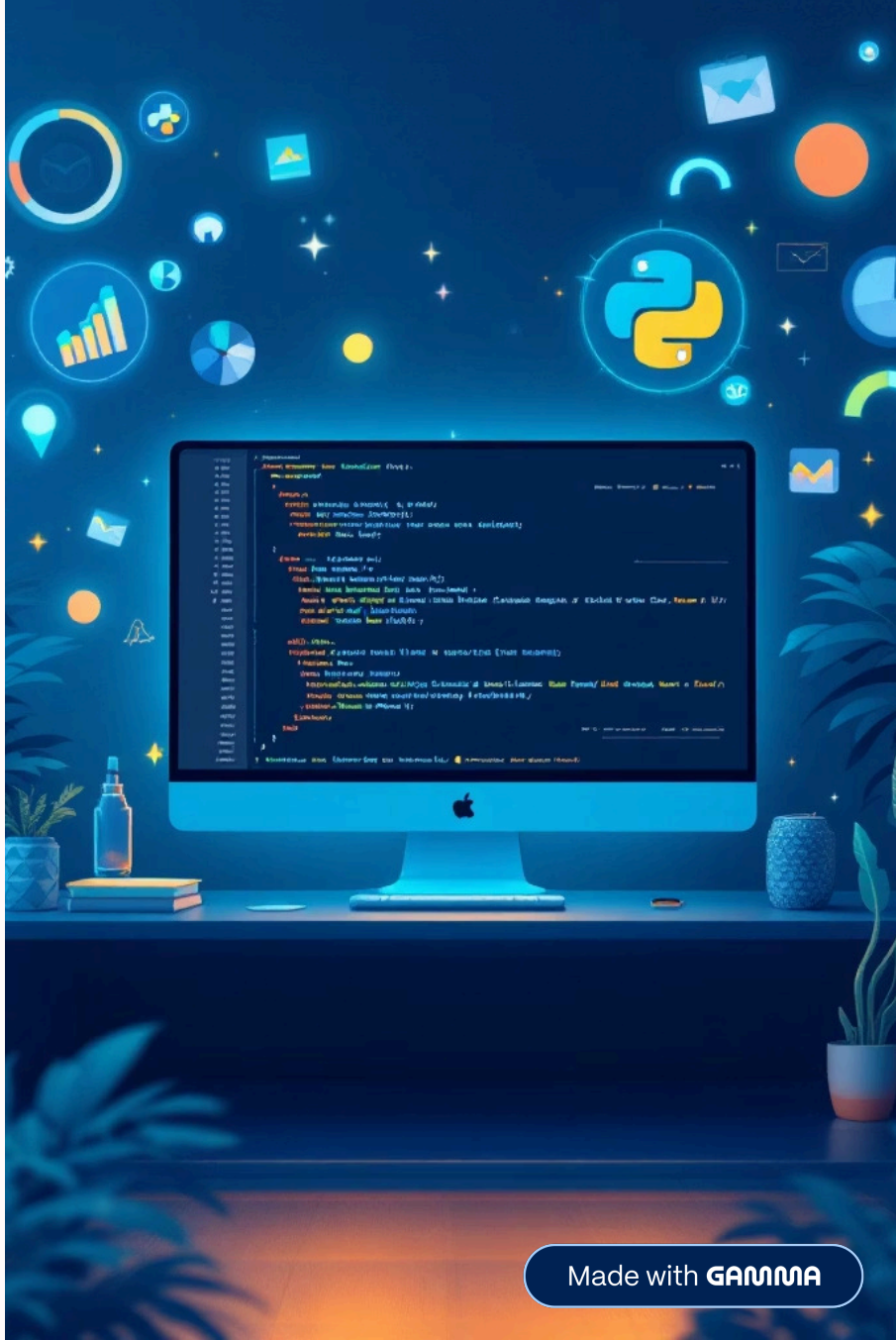

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A Glimpse Ahead

1 Python Basics

Setup, Data Types, Conditionals,

Functions, Loops, Intro to Numpy and Pandas Libraries

2 Data Wrangling and Visualization

Dataframes, Data Cleaning and Manipulation,

Visualizations with Matplotlib and Seaborn Libraries.

3 Probability and Modeling

Review of Probability Theory, What are Models?, Regression, Scikit-learn Library

4 Intro to Machine Learning

Supervised ML, Classification, Regression Again, KNN, Decision Trees, Random Forest

5 Advanced Machine Learning

Unsupervised Learning, K Means Clustering, PCA, NLP and NLTK Library

6 Even More Advanced ML

Deep Learning, TensorFlow Library, Intro to GenAI

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Goals and Practices

01

Hands-On Learning

- Code along with me in Jupyter Notebook
- Engage in fun exercises and data challenges

02

Real-World Data

- Work with real-world datasets to solve actual problems and build your portfolio

03

Build Confidence

- Ask questions that matter
- Analyze data and tell its story in a clear, engaging way

04

Keep Growing

- Gain skills and guidance to continue learning beyond this workshop

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What is Data Science?

Data science is using data to understand the world and make better decisions.

Question

Starting with curiosity, a questions that matter to us. Th best ones come from real-life observations (i.e., data).

Statistics

Providing the mathematical backbone to analyze trends, quantify uncertainty, and make data-driven predictions.

Programming

Using languages like Python or R to clean, manipulate, and model complex datasets.

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Is Mr. Cooper Random?

Time	Time Since Meal	Time Since Last Potty	Activity Before	Location	Outcome
7:45	20 min	1 hr	eating	outside	success
9:10	1 hr 45 min	1 hr 25 min	playing	inside	accident
10:30	2 hr	1 hr 20 min	sleeping	outside	success

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Why Do We Care?

Data is everywhere!

From our earliest moments, data shapes our world:

Early Life

At birth, doctors immediately measure our height, weight, and heart rate.

Growing up

We collect and interpret data constantly: test scores, practice times, and other performance metrics that help us track progress.

Everyday Life

Every interaction on social media, every click and view, generates vast amounts of data.

Making Sense of the World

Data science helps us make sense of all this information by identifying patterns and supporting evidence-based decisions.

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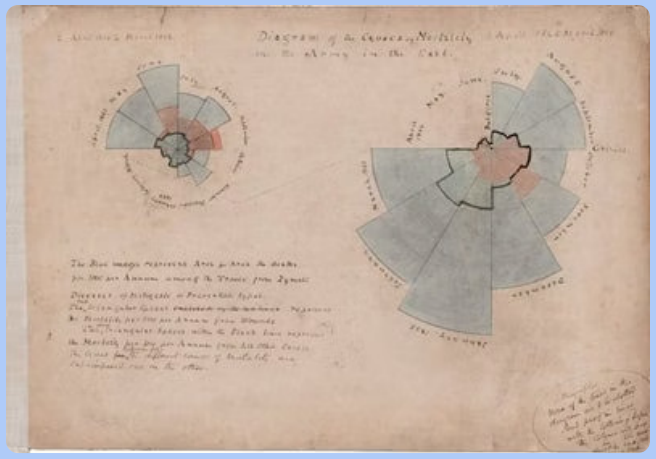
Data Changes Lives

Data science has changed history and continues to shape our future.

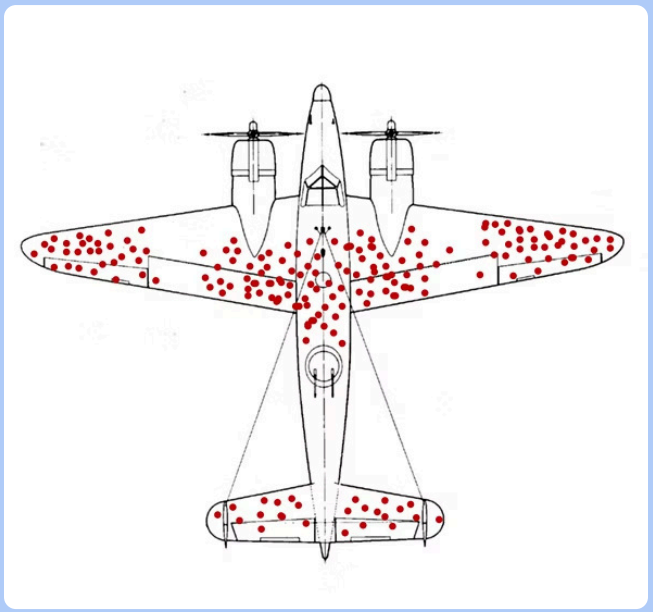
Reforming Public Health

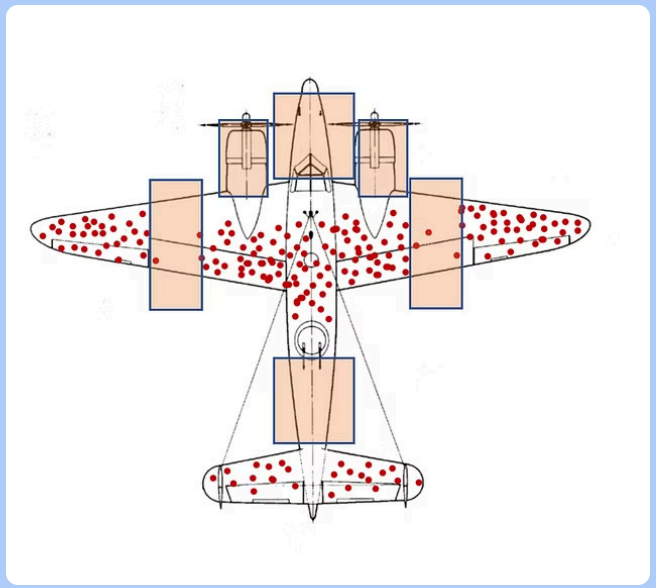
Florence Nightingale's data visualizations revealed unsanitary conditions, leading to public health reforms.



Survivorship Bias

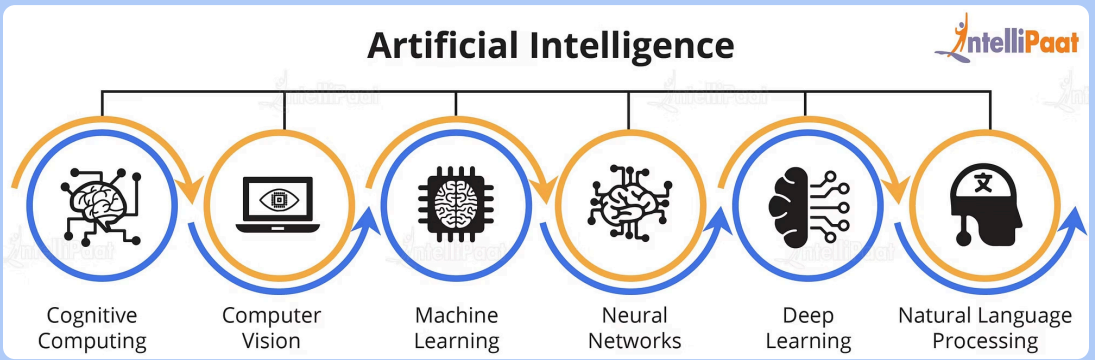
Abraham Wald used data during WWII to improve aircraft survivability by correcting for survivorship bias.





Shaping the Future

Data science is directly giving rise to artificial intelligence, transforming our everyday life.



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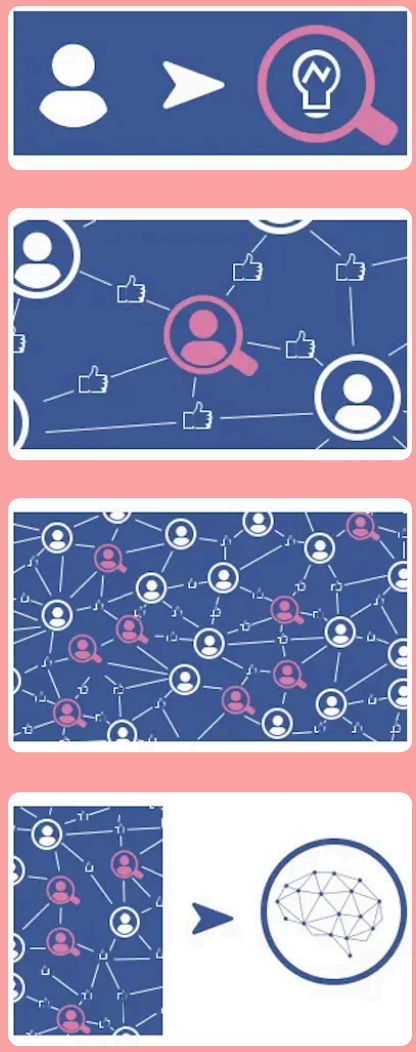
Data Changes Lives

Sometimes for the Worse ...

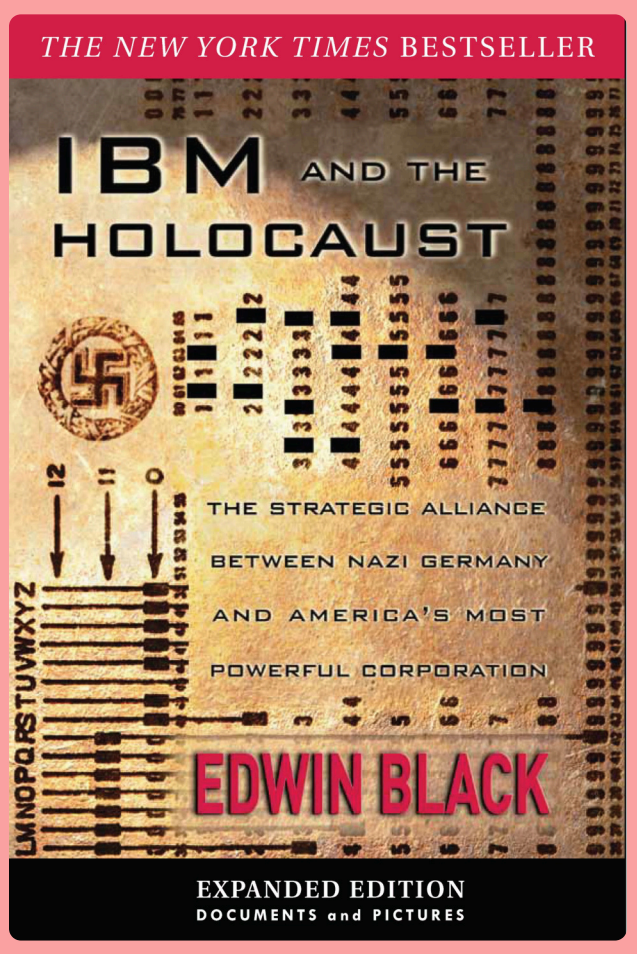
Facebook's week of shame: the Cambridge Analytica fallout

Mark Zuckerberg kept his silence – then did little to assuage the anger in a week that laid bare the worst of Silicon Valley

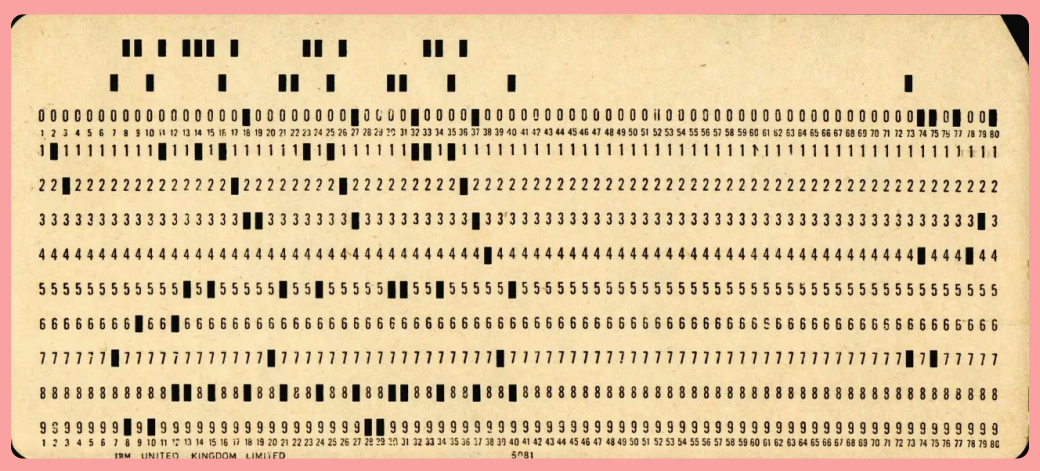
In the 2010s, **personal data** belonging to **millions of Facebook users** was collected by British consulting firm Cambridge Analytica for **political advertising** without informed consent (Image from Guadrian.com).



(Image from BBC.com)



The punch-card system developed by **IBM** allowed the **Nazi regime** to **efficiently categorize and track people**, which made mass persecution easier.



Use Data Responsibly and Ethically to Make the World a Better Place (The Belmont Report).

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Let's get started with Python ...

→ **Versatile & Beginner-Friendly**

→ **Powerful Libraries**

→ **Thriving Community**

Millions of developers and data scientists ready to help you grow.

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